Sr. No	Outlook	temp	Humidity	windy	Ra
1	Sunny	hot	high	weak	
2	Sunny	mild	high	strong	
3	Cloudy	hot	high	weak	
4	Rainy	mild	high	weak	
5	Rainy	cool	normal	weak	
6	rainy	cool	normal	strong	
7	cloudy	mild	normal	strong	
8	Sunny	cool	normal	weak	

Entropy of Prime attribute :-

Rainy → Yes / No → 2 decisions

$$\text{Entropy} = -\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$= -0.5\,\frac{\log_2 0.5}{\log(0.5)/\log(2)} - 0.5\log_2 0$$

$$= -0.5(\quad) - 0.5(\quad)$$

$$= -(\quad + \quad)$$

$$= 1$$

Outlook :-

Sunny :-

$$\text{Entropy} = -\frac{0}{3}\log_2\frac{0}{3} - \frac{3}{3}\log_2\frac{3}{3}$$

$$= 0$$

Cloudy :-

$$\text{Entropy} = -\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}$$

$$= 0$$

Rainy :-

$$\text{Entropy} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= 0.918$$

$$\text{Weighted Entropy} = \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.918)$$

$$= 0.344$$

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Information Gain = Entropy of label - Weighted Entropy of outlook

= 1 - 0.334

= 0.655

Entropy of temp :-

Hot

Entropy = $-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}$

= 1

Cool :-

Entropy = $-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$

= 0.918

Mild :-

$$\text{Entropy} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= 0.918$$

$$\text{Weighted} = \frac{2}{8}(\quad) + \frac{3}{8}(\quad) + \frac{3}{8}(\quad)$$

$$= 0.9385$$

Info Gain = label Entropy - weighted sum

$$= 1 - 0.9385$$

$$= 0.0615$$

Humidity :-

High

$$\text{Entropy} = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}$$

$$= 1$$

Normal

Low :-

$$\text{Entropy} = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}$$

$$= 1$$

Weighted Sum $= \frac{4}{8}(1) + \frac{4}{8}(1)$

$= 1$

Info Gain = Label Entropy - Weight

$= 1 - 1$

$= 0$

Windy :-

Strong

$$\text{Entropy} = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$$

$$= 0.918$$

Weak

$$\text{Entropy} = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}$$

$$= 0.2922$$

Weighted Sum

$$= \frac{3}{8}(0.918) + \frac{5}{8}(0.2922)$$

$$= 0.526$$

Info Gain = Label Entropy - weighted Sum

$= 1 - 0.526$

$= 0.473$

Outlook ... Info Gain